UNIT II

Accident Prevention & Protective Equipment

PPE | plant survey | body parts to protect | training | causes & cost | housekeeping | first aid | reporting | investigation | psychology

I. Personal Protective Equipment (PPE)

PPE is any equipment worn by a worker to protect against one or more hazards that cannot be removed by engineering or administrative controls. It is the **last line of defence** in the hierarchy of control - it does not remove the hazard, it only protects the wearer.

Requirements of good PPE

> Gives **adequate protection** against the specific hazard and meets the **IS / ISO / ANSI** standard.

> Light, comfortable, properly **fitting**; does not hinder movement or vision.

> Durable, easy to clean / maintain / store; reasonably cheap.

> Does not itself create a new hazard; workers are **trained** to select, use, inspect and discard it.

Types of PPE and what they protect

Body part	PPE	Hazard / remark
Head	Safety helmet (hard hat), bump cap, hair net	Falling objects, impact, electric shock (class E), entanglement of hair
Eyes & face	Safety spectacles, goggles, face shield, welding helmet	Flying chips, dust, splashes, UV/IR radiation, sparks
Ears	Ear plugs (20-30 dB), ear muffs, canal caps	Noise above 85-90 dB(A)
Respiratory	Dust mask, half/full-face respirator, SCBA, air-line respirator	Dust, fumes, vapours, toxic gas, oxygen-deficient air
Hands	Gloves - leather, rubber, PVC, cotton, chemical-resistant, heat-resistant	Cuts, burns, chemicals, vibration, electricity
Feet & legs	Safety shoes with steel toe cap, gumboots, leggings, spats	Falling/rolling objects, nails, hot metal, wet floor, chemicals
Body	Overalls, aprons, aluminised suits, acid-proof suits, hi-vis jackets	Heat, splashes, sparks, vehicles, entanglement

Body part	PPE	Hazard / remark
Fall protection	Full-body harness, lanyard, lifeline, safety net	Work at height above 2 m

Respirators – quick classification

Type	Principle	Use
Air-purifying (filter / cartridge)	Cleans the ambient air through filter or chemical cartridge	Normal oxygen (>19.5%), known contaminant at low level
Air-supplying (air-line, SCBA)	Supplies clean breathing air from outside	Oxygen-deficient, unknown or IDLH atmosphere, rescue, tanks

Selection of PPE – 5 steps

> Identify the hazard & the body part exposed (job hazard analysis).
> Check that engineering controls are not possible / not enough.
> Choose PPE with the correct **rating** (e.g. dB reduction, chemical resistance).
> Fit-test and **train** the user; provide replacements & storage.
> Inspect before each use, **supervise** wearing, replace when worn out.

II. Survey of the plant for locations (safety survey)

A **plant safety survey** is a systematic walk-through to **locate hazards** and decide where PPE or other control is needed. It is done by the safety officer with supervisors and workers' representatives.

1. Collect layout, process flow chart, list of machines, materials, chemicals and number of workers.
2. Walk through each department and note **hazard locations**: moving parts, hot surfaces, noise, dust, fumes, work at height, electrical panels, material handling routes, storage.
3. Rate hazards – probability x severity – to **prioritise** (HIRA).
4. Mark the **PPE zones** (e.g. 'Ear protection compulsory', 'Helmet area') on a plant map with safety signs.
5. Decide controls: guard, ventilate, enclose, change process; then PPE for residual risk.
6. Record findings, assign responsibility with target dates, and **re-survey** periodically.

Location in plant	Typical hazard	Protection needed
Forging / foundry	Heat, noise, molten metal	Face shield, aluminised apron, ear muffs, leather shoes
Grinding / machine shop	Flying chips, rotating parts, noise	Goggles, no loose clothes, guards, ear plugs
Welding shop	UV/IR rays, fumes, sparks	Welding helmet, gloves, ventilation, screens
Chemical / battery room	Acid splash, vapours	Acid-proof suit, goggles, respirator, eye-wash station
Store / loading dock	Heavy lifting, forklifts	Safety shoes, hi-vis vest, safe lifting training
Elevated work	Falls	Harness, scaffold tags, edge protection

Parts of the body to be protected – and the injuries

> **Head & brain** – fracture, concussion, burns -> helmet.

> **Eyes** – 'eyes are irreplaceable'; chemical burns, foreign body, blindness -> goggles / face shield.

> **Ears** – noise-induced hearing loss (permanent) -> ear plugs / muffs.

> **Lungs** – pneumoconiosis, silicosis, poisoning -> respirator, ventilation.

> **Hands & fingers** – most frequently injured (cuts, crush, amputation) -> gloves, guards, tools.

> **Feet** – crush, puncture, slips -> safety shoes.

> **Skin** – dermatitis, burns -> aprons, barrier cream.

> **Back** – strain, hernia from lifting -> training, mechanical aids, back support.

III. Education and training in safety

People cause most accidents, so people must be educated. **Training** gives knowledge + skill + attitude to work safely.

Method / stage	Content
Induction training	New workers: plant rules, hazards, emergency exits, PPE, reporting, who to call – before starting work.
On-the-job training	Supervisor teaches the safe method step-by-step (show, explain, let him try, correct).
Tool-box talks	5-10 min talk at start of shift on one topic (e.g. housekeeping, ladders).
Refresher & special	Fire fighting, first aid, confined space, crane signalling, chemical handling – periodically.

Method / stage	Content
Audio-visual & posters	Films, models, demonstrations, posters, slogans, safety notice boards, bulletins.
Campaigns	National Safety Week (4-10 March), safety contests, suggestion schemes, rewards.
Mock drills	Practice of evacuation, fire fighting, gas leak control.

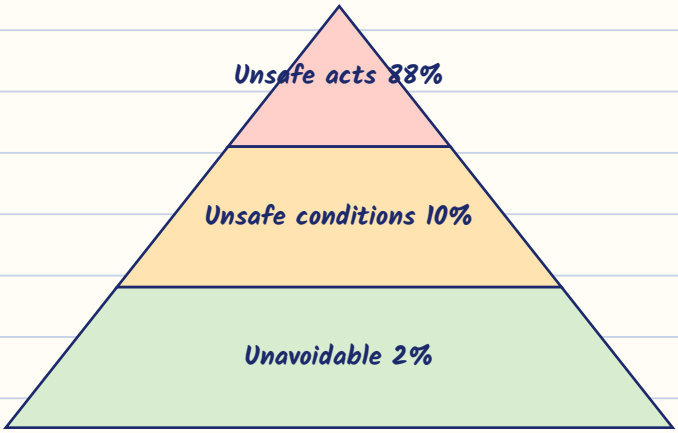
Good training: **Tell** -> **Show** -> **Do** -> **Review**

Training for supervisors is the key

> Supervisor is the **link** between management and workers - he enforces rules every day.

> Train supervisors in **job safety analysis, accident investigation, communication, leadership**.

IV. Causes of accidents



Accident causes

Direct: unsafe act or condition

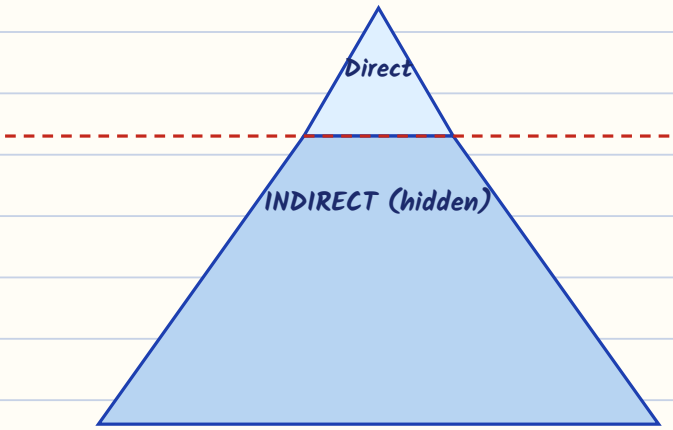
Indirect: management, planning, training, supervision gaps

Root cause analysis asks 'WHY?' five times.

Category	Examples
Unsafe acts	Operating without authority, working at unsafe speed, removing guards/safety devices, not using PPE, using defective tools, unsafe lifting, horseplay, taking shortcuts, working on moving machines, drugs/alcohol.
Unsafe conditions	Unguarded or defective machines, poor housekeeping, inadequate light / ventilation, high noise, congestion, bad layout, improper storage, missing warning signs, hazardous atmosphere.

Category	Examples
Personal factors	Lack of knowledge/skill, poor attitude, fatigue, stress, physical defects (eyesight/hearing), carelessness, alcohol.
Job / management factors	Wrong selection, poor training, inadequate supervision, no safety policy, weak maintenance, pressure on production, poor communication.
Environmental	Heat, humidity, rain, slippery floors, dust, vibration.

V. Cost of accidents



Direct (visible): medical, compensation, repair of damage - about **1 part**

Indirect (hidden): lost time, damaged plant, retraining, lower morale, legal - about **4 parts** (Heinrich)

Direct (insured) costs	Indirect (uninsured, hidden) costs
Medical & hospital expenses	Time lost by injured worker and by co-workers who stop to help / watch
Compensation & disability payments	Damage to machine, tools, material, product; repair time
Insurance premium increase	Supervisor time spent on investigation, reports, arranging replacement
Fines & legal costs	Training cost of replacement; lower output of new/injured returning worker
Funeral / dependants' relief	Loss of orders, bad publicity, lower morale, strikes, overtime

> Heinrich: **indirect cost = about 4 x direct cost**, so total cost is about 5 times the insured cost.

> Cost-benefit: Rs 1 spent on prevention saves several rupees of total accident cost.

> Human suffering of the victim and family cannot be measured in money.

VI. Housekeeping

Housekeeping means keeping the work-place clean, tidy and orderly - 'a place for everything and everything in its place'. Poor housekeeping is the cause of slips, trips, falls, fires, and obstructed escape routes.

5S	Meaning	In plant
Seiri	Sort	Remove unneeded items, scrap, old tools
Seiton	Set in order	Marked places, labels, shadow boards
Seiso	Shine / clean	Daily cleaning, oil-leak repair, waste bins
Seiketsu	Standardise	Checklists, colour codes, procedures
Shitsuke	Sustain	Discipline, audits, habit

> Keep **aisles and exits** free (yellow lines), floors dry and non-slip; clean spills at once.

> Store materials in racks, stacked stable, heavy items low; flammables in approved cabinets.

> Waste: segregated bins - oily rags in covered metal bins (spontaneous ignition), scrap removed daily.

> Good lighting, tool racks, clear signs; pipes colour-coded; machine area clean of oil/ swarf.

> Benefits: fewer accidents & fires, better productivity, space saved, good morale & image.

VII. First aid

First aid is the immediate care given to an injured or suddenly ill person **before** medical help arrives, to **preserve life, prevent worsening, promote recovery**. Law (Sec 45) requires a first-aid box per 150 workers, under a trained person.

Contents of a first-aid box (typical)

> Sterilised dressings, bandages (triangular & roller), adhesive plasters, cotton wool, gauze.

> Antiseptic lotion, burn ointment, eye wash, safety pins, scissors, tweezers, splints.

> Pain relief tablets, ORS, ammonia inhalant; first-aid manual & list of emergency numbers.

Basic first-aid actions

Situation	What to do
Bleeding	Apply direct pressure with clean pad; raise limb; do not remove soaked pad - add more; bandage; shift to hospital.
Burns	Cool with running water 10-20 min, remove rings; cover with clean non-fluffy dressing; no ice, butter, or bursting blisters.
Fracture	Do not move the limb; immobilise with splint above and below; support; treat for shock.

Situation	What to do
Electric shock	Switch OFF supply / push away with dry non-conductor; check breathing; start CPR if needed.
Chemical in eye	Wash with plenty of clean water for at least 15 min, eyelids open; cover; send to doctor.
Unconscious	Check danger, response, airway, breathing; recovery position; CPR (30 compressions : 2 breaths) if not breathing.
Poisoning / gas	Move to fresh air (rescuer with breathing apparatus), loosen clothing, oxygen, medical aid.

Golden rules of first aid

> Remain calm; make the area **safe** first (switch off power, remove from gas).

> Call for medical help early; do not give food/drink to an unconscious victim.

> Record the case in the first-aid register and report if it is an accident.

VIII. Accident reporting

Reporting gives **data** to find causes and fulfils **legal duty**. Every accident, injury, dangerous occurrence and near-miss should be reported, immediately, to the supervisor.

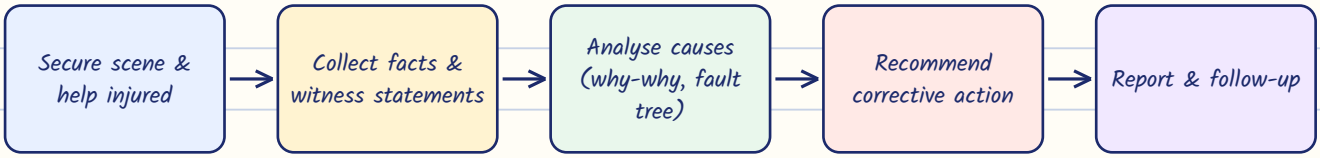
Type of report	Details
Internal report	Accident report form by supervisor within 24 hrs: who, when, where, what, injury, witness, first-aid, cause.
Notice of accident (Sec 88)	Death or injury that disables for 48 hrs or more: notify the Inspector immediately by phone/fax and send a written report (Form 18) in 12 hrs; death - within 4 hrs.
Sec 88A	Reporting of dangerous occurrences (explosion, collapse of crane, bursting of vessel, fire causing stoppage) even if no one is hurt.
Sec 89	Notice of certain occupational diseases (lead poisoning, anthrax, silicosis...) to the Inspector.
Registers	Accident book / register kept by the factory; annual returns (Form 22) to the Inspector.

> Statistics: Frequency rate, severity rate, incidence rate (see Unit I) and cost of accidents are calculated from these reports.

> A **no-blame** reporting culture increases reporting of near-misses; hiding accidents is a punishable offence.

IX. Accident investigation

Purpose: find the **root causes** to prevent repeat - **not to blame**. Start immediately while evidence is fresh.



1. **Preserve the scene**: photograph, sketch, keep failed parts, note time, weather, lighting, machine state.
2. **Interview** the injured, witnesses, supervisor separately - open questions, no pressure.
3. **Review records**: training, maintenance, permits, SOP, previous incidents.
4. **Analyse**: chain of events; 5-Why; fishbone (Ishikawa); fault tree; separate immediate / underlying / root causes.
5. **Recommend** actions with owner & date: engineering fix first, then procedure, then training.
6. **Follow-up**: verify implementation; share lessons plant-wide; update risk assessment.

Investigation team

> Supervisor of the area, safety officer, a worker representative, maintenance/engineer, and for serious cases a senior manager and external expert.

X. Industrial psychology in accident prevention

Industrial psychology studies human behaviour at work - how perception, attitude, motivation, fatigue and stress affect accidents.

Factor	Effect on accidents / remedy
Accident proneness	Some individuals have more accidents than others due to personality (impulsive, careless, risk-taker) or physical limits; remedy: correct selection & placement, counselling, job change.

Factor	Effect on accidents / remedy
Attitude & motivation	Negative attitude -> shortcuts. Motivate by participation, recognition, rewards (Maslow's hierarchy - safety is a basic need).
Fatigue	Physical or mental tiredness reduces attention, reaction; remedy: rest pauses, proper shifts, good posture, less overtime.
Monotony & boredom	Repetitive work lowers alertness; remedy: job rotation, enrichment, music, breaks.
Stress & emotional state	Worry, anger, family problems cause lapses; remedy: counselling, support, stress management.
Perception & skill	Poor vision / hearing, wrong judgement; remedy: medical checks, training, ergonomic design.
Group behaviour	Peer pressure and leadership influence safe/unsafe norms; remedy: team safety meetings and role models.

> **Selection tests**: aptitude, vision, reaction time to place right person on right job.

> **Safety attitude change**: persuasion, role-play, films, discussion; positive reinforcement works better than punishment.

XI. Safety trials & safety audits

Safety trials are planned checks to prove that a machine, process or system is safe before routine use, and to keep it safe afterwards.

Check	Purpose / how
Trial run / commissioning	New or modified machine run at no-load and load with safety devices tested, by a competent person, before acceptance.
Safety inspection / tour	Planned walk-through with checklist - daily (supervisor), weekly (department), monthly (committee), annual (management).
Safety sampling	Random observation of unsafe acts/ conditions in a sample of workers; computes % unsafe, to find trends.
Safety audit	Independent, systematic examination of the safety system against standards; gives scores and corrective-action plan.
Safety survey	Detailed study of one problem area (noise, dust, fire) using instruments.

Check	Purpose / how
Hazard & operability study	HAZOP team applies guide words (no, more, less, reverse) to process lines to find deviations.

XII. Safety related to operations

Operation	Key safety measures
Machine operation	Guards (fixed, interlocked, trip), emergency stop, no loose clothing/jewellery, switch-off & **lock-out/tag-out** before maintenance.
Material handling	Correct lifting posture (bend knees, back straight), mechanical aids, SWL marked on cranes, tested slings, no standing under load.
Work at height	Harness, scaffold inspection, guard-rails, ladders at 4:1 angle, permit for work above 2 m.
Hot work (welding/cutting)	Hot-work permit, remove flammables, fire extinguisher & fire watch, gas cylinders upright & chained.
Confined space	Entry permit, isolate, test oxygen (19.5–23.5%) & toxic gas, ventilate, standby person, rescue plan.
Electrical work	Isolate and earth, insulated tools, ELCB/RCCB, do not wet-handle, competent persons only.
Chemical handling	Labels & SDS, correct storage & segregation, PPE, eye-wash/shower, spill kit.
Pressure systems	Safety valves, periodic testing, no over-pressure, statutory inspection.

Permit-to-Work + Lock-Out/Tag-Out = control of dangerous operations

XII-A. Additional notes (important for exams)

Machine guarding – types of guards

Guard type	Working	Example
Fixed guard	Permanent barrier; tool needed to remove	Gear cover, belt enclosure
Interlocked guard	Machine cannot run unless guard is closed	Press-brake door, centrifuge lid
Adjustable guard	Adjusted to suit the job size	Band-saw blade guard, drill guard

Guard type	Working	Example
Self-adjusting guard	Moves as stock is fed	Circular saw blade guard
Trip / pull-back device	Stops the machine if hand enters danger zone	Light curtain, trip bar
Two-hand control	Both hands kept away at start of cycle	Power press

Lock-out / Tag-out (LOTO) – 6 steps

1. Inform affected persons and identify all energy sources (electric, hydraulic, pneumatic, gravity, thermal).
2. Shut down the machine in the normal way.
3. **Isolate** the energy source (open switch / valve).
4. **Lock & tag** the isolating point – each worker puts his own padlock; tag shows name, date.
5. Release stored energy (bleed, ground, block), then **verify zero energy** by trying to start.
6. After work: remove tools, refit guards, clear people, remove own lock, re-energise.

Safety colour code & signs (IS 9457)

Colour	Meaning	Example
Red	Prohibition / fire equipment / stop	No smoking, fire extinguisher
Blue	Mandatory action	Wear helmet, ear protection
Green	Safe condition, first-aid, exit	Emergency exit, first-aid post
Yellow	Warning / caution	Danger – high voltage, wet floor

Safe manual lifting

> Plan the lift, check the load and path; get help or a trolley if load > 25 kg (for men; less for women).
> Feet apart, bend knees, **back straight**, firm grip, lift with legs, keep load close to body.
> Do not twist while carrying; turn with the feet; put down in the reverse way.

Hand-tool safety

> Use the right tool for the job; hammer heads wedged, mushroomed chisels dressed.
> Spanners not extended with pipes; files with handles; screwdriver not used as chisel.
> Tools inspected, stored in racks; carried in a bag when climbing; power tools earthed and double-insulated.

Fault-tree analysis (FTA) – tool for accident analysis

FTA is a top-down, logic-diagram method: start with the **top event** (accident), then find combinations of causes joined by **AND** gates (all must occur) and **OR** gates (any one is enough) down to basic events. Probability of the top event can be calculated; the diagram shows the weakest links to strengthen.

Gate	Output occurs when...	Probability
AND	All input events occur	P = P1 x P2 x ...
OR	Any input event occurs	P = 1 - (1-P1)(1-P2)... = P1 + P2 - P1P2

Eg: fire occurs if (fuel AND oxygen AND ignition). Fuel leak P=0.1, ignition source P=0.05, oxygen is always present (1) -> P(fire) = 0.1 x 0.05 = **0.005**.

Contents of a permit-to-work

> Job description & location, date/time validity, who will do it.

> Hazards identified and precautions (isolation, gas testing, PPE, fire watch, rescue).

> Signature of issuer (authorised person), acceptor, and a final **cancel/close-out** signature.

XIII. Quick revision & important questions

Remember

> PPE = last line of defence; Hierarchy: Eliminate > Substitute > Engineering > Admin > PPE.

> Heinrich: indirect cost = 4 x direct; 88% unsafe acts.

> Accident notice: injury disabling 48 hrs / more (Sec 88), dangerous occurrence (88A).

> First aid box: 1 per 150 workers (Sec 45).

> Investigation aims at root cause, not blame.

> 5S = Sort, Set in order, Shine, Standardise, Sustain.

1. Explain the types of PPE and the parts of the body they protect. (10 M)
2. How is a plant surveyed for safety? Discuss. (5 M)
3. Discuss the causes and cost of accidents. (10 M)
4. Write about housekeeping and 5S. (5 M)
5. Describe first-aid for burns, bleeding and fractures. (5 M)
6. Explain the procedure of accident reporting and investigation. (10 M)
7. Role of industrial psychology in accident prevention. (10 M)